

August 24, 2026

Dear Yelp Hiring Team,

I am writing to apply for the Entry Level Machine Learning Engineer position at Yelp. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python/C++ systems for machine learning, data pipelines, model evaluation, and backend APIs, I am excited by the chance to help Yelp turn large-scale user and business data into useful ML signals and GenAI applications.

What draws me to this role is the end-to-end nature of the work. Yelp describes a role that spans raw data, model development, pipelines, deployment, and practical tradeoffs around scale, latency, cost, and availability. My graduate research required a similar habit of thinking beyond a model alone. I built Python and C++ workflows for ingesting, preprocessing, structuring, retrieving, and evaluating large Earth observation datasets, including DGGS-based pipelines that convert imagery into structured tensor data. I also improved one C++ processing workflow from 233 seconds to 26 seconds for a 10-tensor workload through multithreading and performance-focused implementation.

I also bring direct ML experience from my DEM super-resolution work, where I trained PyTorch ResNet models, built data collection and preprocessing workflows using Google Earth Engine and Rasterio, and evaluated model quality with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. That project made me especially careful about data quality, validation strategy, model evaluation, and making model behavior understandable through visualization tools.

Beyond research, I have built backend services with Django REST APIs, PostgreSQL, authentication flows, payment-service integration, and a WebRTC-based chatbot system using GPT-style responses and FAQ data. I have also been building an agentic job-search automation platform with LLM-based workflows, Google Sheets and Drive integration, Telegram alerts, n8n, OpenClaw, Docker, and SearXNG. These projects connect well with Yelp's interest in LLMs, agentic design, ML pipelines, and systems that are useful to real users.

I have not yet worked professionally with Spark MLlib, Redshift, Cassandra, Kafka, or Flink, but I bring strong Python and C++ foundations, PyTorch and scikit-learn experience, backend/API development, SQL, Docker/AWS exposure, and a genuine appetite for debugging systems deeply and systematically. I would be excited to grow in Yelp's collaborative engineering environment while contributing reliable code to ML systems that connect people with local businesses across Canada and beyond.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in ML workflows, scalable data processing, performance optimization, backend development, and AI automation can contribute to Yelp's machine learning engineering team.

Sincerely,  
Amir Mirzai Golpayegani  
amirgolpaa24@gmail.com